

Admissible Observable Sectors in Optical Measurement Systems

Fisher Resolution, Stable Null Directions, and Low-Dimensional Transport

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Abstract

We formulate a measurement-side version of Resolution Geometry (RG) in which experimentally accessible structure is represented by an admissible observable sector $W_{\text{obs|adm}}$, rather than by raw algebraic rank or by an ontological claim about the measured system. Candidate directions are first identified through the singular or Fisher spectrum of a measurement operator and are then subjected to rank-stability, bounded-conditioning, coarse-graining, and transport tests. The framework is evaluated on public undergraduate optics data comprising photodiode frequency response under five load conditions, six-channel LED spectroscopy, optical activity, polarization, and single- and double-slit diffraction. The strongest numerical results are: stable load-dependent cutoff frequencies spanning more than two orders of magnitude; a cross-load response atlas of 99% energy rank two; a stable five-dimensional 99% spectral sector with a sixth direction carrying approximately 0.29% of normalized energy; and an exact two-dimensional null space for the uncontrolled optical-activity product model $\alpha = kLc$, removed only by external experimental controls. These results support RG as a geometry of experimental accessibility and apparatus-dependent sector transport. They do not establish that observation causes interference loss, that path knowledge alters the ontology of light, or that RG replaces quantum complementarity. A limited double-slit connection is stated as a falsifiable conjecture requiring joint measurements of path distinguishability and visibility under a common intervention.

1 Introduction

A measurement does not expose every physical or model parameter with equal stability. Some directions are strongly identifiable, some are weakly conditioned, and some are exactly or effectively null. This motivates a distinction between the raw observable space and the subset that remains usable after experimental admissibility requirements are imposed.

The purpose of this paper is deliberately limited. We do not assign wave-like or particle-like ontology to light. Instead, we ask whether a family of measurement operators admits a coordinate-covariant, noise-aware, stable decomposition into resolved and null directions, and whether those decompositions can be transported between experimental configurations.

The motivating double-slit question is correspondingly modest. If a path-marking interaction is varied, one may ask whether a path-distinguishing direction enters the admissible observable sector while a mutual-coherence direction leaves it. The present data do not contain the joint intervention required to test that exchange. They do, however, allow the underlying measurement-side construction to be attacked independently across several optical systems.

Contributions. The paper provides:

- (i) a definition of $W_{\text{obs|adm}}$ based on Fisher resolution followed by explicit admissibility gates;
- (ii) a theorem ladder establishing coordinate covariance, local sector constancy, monotonicity under deterministic coarse-graining, and bounded transport of retained directions;

- (iii) real-data demonstrations of stable resolution boundaries, stable weak directions, controlled removal of exact degeneracies, and a low-dimensional common atlas across apparatus settings;
- (iv) explicit non-claims separating demonstrated measurement geometry from untested claims about complementarity or causation.

2 Measurement model and admissible sectors

Let Θ be a finite-dimensional parameter manifold, $\mathcal{O}_g$ an observation space indexed by apparatus configuration g , and

$$y_g = A_g(\theta) + \varepsilon_g, \quad \mathbb{E}[\varepsilon_g] = 0, \quad \text{Cov}(\varepsilon_g) = C_g(\theta). \quad (1)$$

For the local Gaussian approximation, let

$$J_g(\theta) = D_\theta A_g(\theta), \quad F_g(\theta) = J_g(\theta)^\top C_g(\theta)^{-1} J_g(\theta). \quad (2)$$

The same construction may be made from a whitened linear operator $C_g^{-1/2} A_g$ when the model is linear.

Definition 2.1 (Candidate resolved sector). *For a fixed Fisher threshold $\tau_F > 0$, define*

$$\widetilde{W}_g(\theta; \tau_F) = \text{span}\{v_i : \mu_i(F_g(\theta)) > \tau_F\}, \quad (3)$$

where μ_i and v_i are eigenvalues and orthonormal eigenvectors of F_g in a chosen local metric.

A Fisher threshold alone is insufficient: a direction can be numerically nonzero yet unstable under resampling, badly conditioned, nontransportable, or generated by nuisance structure.

Definition 2.2 (Admissibility gates). *A candidate subspace $W \subseteq T_\theta \Theta$ is admissible at (g, θ) if it satisfies all declared gates:*

- (A1) rank stability: *its dimension and principal angles are stable under the specified resampling or perturbation class;*
- (A2) bounded conditioning: $\text{cond}(F_g|_W) \leq \kappa_{\max}$;
- (A3) coarse-graining compatibility: *allowed observation maps do not create information relative to the declared statistical model;*
- (A4) transportability: *the comparison map between overlapping experimental charts is bounded on W ;*
- (A5) nuisance closure: *declared gauge and nuisance directions have been quotiented out or projected away.*

Thresholds and perturbation classes are part of the experimental specification, not universal constants.

Definition 2.3 (Admissible observable and null sectors). *Let $\Pi_{\text{adm},g}$ denote the composite admissibility selection. Define*

$$W_{\text{obs|adm}}(g, \theta) = \Pi_{\text{adm},g} \widetilde{W}_g, \quad N_{\text{adm}}(g, \theta) = W_{\text{obs|adm}}(g, \theta)^{\perp_F}, \quad (4)$$

where the complement is taken after nuisance elimination and relative to the retained Fisher geometry whenever it is nondegenerate.

Remark 2.4. $W_{\text{obs|adm}}$ is protocol-relative. Changing detector load, bandwidth, wavelength sampling, controlled variables, noise floor, or admissibility threshold may change the sector without requiring any change in the ontology of the source.

3 Theorem ladder

3.1 Coordinate covariance

Theorem 3.1 (Covariance of Fisher support). *Let $\eta = \varphi(\theta)$ be a smooth local reparameterization with invertible derivative $B = D_\eta\theta$. Then*

$$F_\eta = B^\top F_\theta B. \quad (5)$$

Consequently, Fisher rank, nullity, and the distinction between identifiable and null tangent directions are invariant under regular coordinate changes.

Proof. From $J_\eta = J_\theta B$ and (2), $F_\eta = B^\top J_\theta^\top C^{-1} J_\theta B = B^\top F_\theta B$. Congruence by an invertible matrix preserves rank and nullity. A tangent vector u is null in η coordinates exactly when Bu is null in θ coordinates. $\square$

Corollary 3.2 (Coordinate-free resolved quotient). *The quotient $T_\theta\Theta/\ker F_g$ is invariant up to canonical local isomorphism under regular reparameterization. Numerical thresholds require a metric and scale convention, but exact Fisher holes do not.*

3.2 Local constancy and admissibility boundaries

Theorem 3.3 (Local constancy away from boundaries). *Assume $g \mapsto F_g$ is continuous and symmetric, and all non-spectral admissibility statistics are continuous on an interval I . If no Fisher eigenvalue equals τ_F and no admissibility statistic equals its boundary at $g_0 \in I$, then there is a neighborhood U of g_0 on which $\dim W_{\text{obs}|\text{adm}}(g)$ is constant.*

Proof. Eigenvalues of a continuous symmetric matrix family are continuous. Strict inequalities $\mu_i(F_{g_0}) \gtrless \tau_F$ persist in a sufficiently small neighborhood. The same holds for each continuous admissibility statistic separated strictly from its threshold. Hence no candidate direction enters or leaves and no retained direction fails or regains a gate on that neighborhood. $\square$

Corollary 3.4 (Resolution transitions). *A change in admissible sector dimension can occur only at a declared boundary, including*

$$\mu_i(F_g) = \tau_F, \quad \text{cond}(F_g|_W) = \kappa_{\max}, \quad s_i(g) = \tau_{\text{stab}}, \quad (6)$$

or at a failure of nuisance closure or bounded transport.

Remark 3.5. This is a change of an admissible stratification, not necessarily a topological change of the underlying physical state space.

3.3 Coarse-graining

Theorem 3.6 (Fisher monotonicity under parameter-independent coarse-graining). *Let Y have density $p_\theta(y)$ and let Z be obtained from Y through a Markov kernel $K(z|y)$ independent of θ . Under the usual regularity conditions,*

$$F_Z(\theta) \preceq F_Y(\theta). \quad (7)$$

Thus deterministic or stochastic post-processing cannot create Fisher information.

Proof. The score of Z is the conditional expectation of the score of Y given Z . For every vector a , conditional Jensen gives

$$a^\top F_Z a = \mathbb{E} \left[\left(\mathbb{E}[a^\top s_Y | Z] \right)^2 \right] \leq \mathbb{E}[(a^\top s_Y)^2] = a^\top F_Y a. \quad (8)$$

Hence the Loewner inequality follows. $\square$

Corollary 3.7 (No resolved-sector creation by coarse-graining). *If the same metric and absolute threshold are used before and after coarse-graining, a direction below threshold cannot be promoted above threshold by parameter-independent processing alone.*

3.4 Transport between measurement charts

Definition 3.8 (Admissible transport). *For overlapping configurations g, h , an admissible transport is a bounded map*

$$T_{g \rightarrow h} : W_{\text{obs|adm}}(g) \longrightarrow W_{\text{obs|adm}}(h) \quad (9)$$

that preserves the declared observational equivalence classes and is compatible with nuisance quotients.

Proposition 3.9 (Conditioned transport of observable directions). *Suppose $T_{g \rightarrow h}$ is injective on $W \subseteq W_{\text{obs|adm}}(g)$ and*

$$0 < m \leq \sigma_{\min}(T_{g \rightarrow h}|_W) \leq \sigma_{\max}(T_{g \rightarrow h}|_W) \leq M < \infty. \quad (10)$$

Then $\dim T_{g \rightarrow h}(W) = \dim W$, and perturbations are amplified by at most M while inverse reconstruction on the image is amplified by at most m^{-1} .

Proof. The positive lower singular bound implies injectivity and a bounded inverse on the image. The upper bound gives forward stability. Rank preservation follows from injectivity in finite dimension. $\square$

Corollary 3.10 (Atlas interpretation). *A family of apparatus-dependent sectors forms a numerically usable observable atlas on overlaps where admissible transports satisfy uniform nonzero lower and finite upper singular bounds.*

3.5 Controlled identifiability

Proposition 3.11 (Control-induced quotient reduction). *For the optical-activity model*

$$\alpha = kLc, \quad (11)$$

one scalar observation with all three quantities unknown has Jacobian

$$J = (Lc, kc, kL), \quad (12)$$

so, away from the trivial zero point,

$$\text{rank } J = 1, \quad \text{nullity } J = 2. \quad (13)$$

If L and c are externally fixed and nonzero, the remaining parameter k is locally identifiable.

Proof. A nonzero 1×3 matrix has rank one and nullity two. With L, c fixed, $\partial\alpha/\partial k = Lc \neq 0$, so the one-parameter Fisher information is positive for finite nonzero noise variance. $\square$

Corollary 3.12. *Resolution belongs to the complete measurement arrangement. It is not an intrinsic label attached to k , L , or c separately.*

4 Data and reproducible attack

The numerical study uses the public *UniPV Physics Laboratory 3* collection (B.Sc. Physics, University of Pavia, academic year 2022–23). The analyzed files comprise five LED–photodiode AC response curves, a six-channel optical spectrum, optical-activity measurements, polarization measurements, and single- and double-slit diffraction tables.

All stochastic calculations used deterministic seed

$$20260802. \tag{14}$$

The principal machine-readable output is `rg.wobs_adm_optics_attack.json`. The earlier descriptive fit output is retained as `rg_optics_battery_results.json`.

4.1 AC response model

Each load condition was fitted with the phenomenological transfer function

$$V(f) = \frac{V_0}{\sqrt{1 + (f/f_c)^{2n}}}. \tag{15}$$

Residual bootstrap resampling with 500 realizations per load tested stability of f_c . This model summarizes the observable response; it is not asserted to be a complete circuit derivation.

4.2 Spectral operator

After subtracting a low-percentile baseline and normalizing each channel, the wavelength-by-channel matrix X was decomposed as

$$X = U\Sigma V^\top. \tag{16}$$

One thousand empirical baseline-noise perturbations tested the stability of the energy-ranked sector.

4.3 Optical activity

The length and concentration series were fitted linearly and bootstrapped 2,000 times. The controlled fits were compared with the exact rank of the unconstrained product model.

5 Results

5.1 Stable apparatus-dependent frequency sectors

Table 1: AC response fits and bootstrap stability.

Load (Ω)	$\hat{f}_c$	Bootstrap 95% interval	SD($\log f_c$)
0	13.16 kHz	12.42 kHz to 13.86 kHz	0.0290
500	1.465 MHz	1.386 MHz to 1.544 MHz	0.0277
1000	508.98 kHz	469.48 kHz to 551.74 kHz	0.0424
5100	56.13 kHz	54.10 kHz to 58.64 kHz	0.0205
10000	35.31 kHz	34.13 kHz to 36.73 kHz	0.0189

All 2,500 bootstrap fits converged. The cutoff boundary moved by more than two orders of magnitude across configurations, whereas each bootstrap interval was narrow relative to

the between-configuration separation. Therefore there exist frequency bands retained in one apparatus sector and rejected in another under the same declared response threshold.

For the relative information proxy $\mu_g(f) = |H_g(f)|^2$, the retained fraction of sampled frequencies ranged from 0.40 to 0.68 at threshold $\mu > 0.5$, and from 0.74 to 1.00 at threshold $\mu > 0.1$. Absolute sector dimensions remain threshold-dependent, but the movement of the boundary is not confined to one finely tuned threshold.

5.2 Cross-load observable atlas

The five normalized response curves were evaluated on a common logarithmic frequency grid. The singular-energy distribution of the resulting atlas matrix was

$$(0.92664, 0.06753, 0.00389, 0.00152, 0.00042). \quad (17)$$

Thus

$$\text{rank}_{99\%}(\mathcal{T}) = 2, \quad \text{cond}(\mathcal{T}) \approx 47.0. \quad (18)$$

Table 2: Cross-load transport spectrum.

Mode	Singular value	Energy fraction
1	27.4214	92.664%
2	7.4028	6.753%
3	1.7764	0.389%
4	1.1108	0.152%
5	0.5830	0.042%

Although five configurations were measured, 99% of their normalized response energy lies in two atlas directions. This supports low-dimensional transport among apparatus patches. The numerical result does not by itself identify the physical meaning or uniqueness of those two coordinates.

5.3 Stable spectral resolved/null split

The spectral singular-energy fractions were

$$(45.562, 22.694, 16.665, 10.297, 4.490, 0.292)\%. \quad (19)$$

Every one of 1,000 perturbations produced

$$\text{rank}_{99\%}(X) = 5. \quad (20)$$

The weakest energy fraction remained within

$$[0.254, 0.325]\%, \quad (21)$$

and was below 1% in every realization.

Accordingly, a 99% energy convention yields a stable five-dimensional admissible sector and one effective null direction. This is not an assertion that the sixth algebraic direction does not exist; at a 99.9% retention criterion it must be retained.

5.4 Optical activity and control-dependent closure

Both controlled directions are stable. In contrast, the unconstrained model has rank one and nullity two for (k, L, c) . The dataset therefore demonstrates both sides of the RG construction: exact null structure before control closure, and a stable resolved coordinate after the controlled variables are fixed.

Table 3: Spectral-sector stability.

Quantity	Result
Base 99% energy rank	5
Perturbed realizations	1,000
Realizations with 99% rank 5	1,000
Weakest-mode base energy	0.292%
Weakest-mode 95% interval	0.254–0.325%
Probability weakest mode < 1%	1.000

Table 4: Bootstrap stability of controlled optical-activity directions.

Series	Slope	Bootstrap 95% interval	Positive fraction
Length	2.3561	2.1952–2.5149	1.000
Concentration	1.0200	1.0020–1.0380	1.000

5.5 Supporting optical observations

The Malus-law dataset supplies a complementary local example: the global angular dependence is strongly fitted, while the local derivative of ideal intensity with respect to angle vanishes at extrema. These points are local Fisher holes for infinitesimal angle estimation under an additive-noise model, not a global disappearance of the polarization angle. Single-slit datasets resolve principally an aperture-width coordinate, whereas double-slit geometry introduces a second estimable geometric coordinate. These observations motivate the sector language but are not used as evidence for path marking or quantum complementarity.

6 Limited connection to the double slit

For a two-path signal, write the observed intensity schematically as

$$I(x; g) = I_1(x; g) + I_2(x; g) + 2 \operatorname{Re}[\Gamma_{12}(g)u_1(x; g)u_2(x; g)^*], \quad (22)$$

where g is a path-marker coupling and Γ_{12} is mutual coherence. Visibility $V(g)$ is determined by the interference term, while path distinguishability $D(g)$ requires a path-sensitive measurement channel.

Definition 6.1 (Coupled sector-exchange hypothesis). *A path-marking experiment exhibits an RG coupled sector exchange if, under a common intervention g ,*

$$\begin{aligned} \mu_{\text{path}}(g) &\text{ increases through an admissibility boundary,} \\ \mu_{\text{coh}}(g) &\text{ decreases through an admissibility boundary,} \end{aligned} \quad (23)$$

with both directions surviving nuisance, stability, conditioning, and transport gates.

Proposition 6.2 (Identifiability requirement for testing sector exchange). *A dataset cannot identify the hypothesis in (23) unless it contains, under a shared intervention index g , observations sufficient to estimate both a path-distinguishing statistic and a coherence or visibility statistic, together with a noise or repetition model.*

Proof. Without path-sensitive observations, $\mu_{\text{path}}(g)$ is not estimable. Without fringe-resolving observations, $\mu_{\text{coh}}(g)$ or $V(g)$ is not estimable. Without a common g , their trajectories cannot be paired. Without repeated observations or an externally justified noise model, Fisher normalization and stability cannot be evaluated. Hence the coupled claim is non-identifiable. $\square$

The present repository lacks all four joint elements: a varied path-marker coupling, a visibility series, a distinguishability series, and repeated detector-level observations under the same intervention. Current support therefore stops at the general apparatus-to-sector map

$$g \mapsto A_g \mapsto W_{\text{obs|adm}}(g), \quad (24)$$

and at low-dimensional transport among optical measurement configurations.

Even a future observation of $D(g)$ increasing while $V(g)$ decreases would not, by itself, prove that an observer’s knowledge causes interference loss. Both quantities may be consequences of the physical coupling g . A causal claim would require an intervention and mediation analysis distinguishing

$$g \rightarrow \Gamma_{12} \rightarrow V \quad (25)$$

from a semantic claim that abstract knowledge alters the physical system.

7 Discussion

7.1 What is supported

The results support four linked statements.

First, optical measurement configurations induce different stable observable sectors. The AC cutoff movement remains well separated under bootstrap perturbation. Second, those sectors need not be unrelated: the five-load response family admits a two-dimensional 99% atlas. Third, effective null directions can be stable rather than numerical accidents, as shown by the spectral bootstrap. Fourth, experimental controls alter the quotient being inferred, as shown exactly by the optical-activity product model.

These are stronger than ordinary curve-fit observations because the focus is not only on parameter values. The construction asks which directions survive a declared chain of admissibility tests and whether they can be compared across charts.

7.2 What remains conventional

Several mathematical ingredients are standard: Fisher information, singular-value decomposition, eigenvalue continuity, the data-processing inequality, bootstrap stability, and condition-number control. RG’s proposed contribution is their closure into a single observable-sector ontology for inference: nuisance elimination, resolved/null decomposition, admissibility gates, and transport between experimental patches. The paper therefore claims an integrated framework and a successful cross-experiment demonstration, not invention of the constituent linear-algebraic facts.

7.3 Threshold dependence

Effective nullity is protocol-dependent. A 99% energy truncation and a 99.9% truncation answer different experimental questions. This is not a defect provided that thresholds are reported before interpretation, sensitivity analyses are performed, and conclusions are not promoted from effective to exact rank without proof.

7.4 Relation to broader RG applications

The same architecture can be used in signals such as ECG: raw variation is first separated from stable, transportable observable structure. The optics results do not derive validity from another domain, but they show that the architecture is not tied to one data type. A joint cross-domain paper would require harmonized admissibility definitions and independently reproducible datasets; it should not infer physical equivalence merely from shared mathematics.

8 Non-claims and limitations

Non-claim 8.1 (No ontology of light). The results do not show that a photon changes from a wave to a particle, or that either description is an intrinsic state selected by $W_{\text{obs|adm}}$.

Non-claim 8.2 (No derivation or replacement of quantum mechanics). No quantum dynamics, Born rule, state-update law, or complementarity inequality is derived. RG is used here as a measurement-side geometric organization.

Non-claim 8.3 (No observational causation of interference loss). The data do not show that acquiring path information causes a slit-pattern change. No path-marker intervention is present.

Non-claim 8.4 (No tested path-coherence exchange). The coupled transition in (23) is a falsifiable hypothesis, not an empirical result of this paper.

Non-claim 8.5 (No threshold-free physical rank). The effective dimensions reported for spectral and transport matrices depend on explicit energy criteria. Only the optical-activity product degeneracy is exact within its stated model.

Additional limitations. The AC transfer function is phenomenological; measurement quantization and heteroscedastic noise were not separately calibrated. The spectral perturbation model is estimated from low-signal values rather than independent dark frames. The optical datasets are small teaching-laboratory samples. The cross-load atlas was evaluated through normalized response curves, so overall gain information was intentionally removed. These choices make the present work a stringent proof of construction and stability, not a precision metrology study.

9 Falsification criteria and next experiment

The measurement-side framework would be weakened if:

- (F1) inferred sectors changed arbitrarily under regular reparameterization after appropriate metric normalization;
- (F2) candidate directions failed resampling stability or required unbounded inverse transport;
- (F3) deterministic coarse-graining systematically created Fisher information under a correctly specified model;
- (F4) purported common-atlas coordinates failed on held-out configurations;
- (F5) sector conclusions depended on an undisclosed, isolated threshold choice.

A direct test of the double-slit hypothesis requires a dataset with apparatus coupling values $g_1, \dots, g_m$, repeated fringe-resolving detector frames, and path-sensitive observations sufficient to estimate $D(g)$. The primary preregistered quantities should be

$$V(g), \quad D(g), \quad \mu_{\text{coh}}(g), \quad \mu_{\text{path}}(g), \quad (26)$$

plus principal-angle stability and transport conditioning between adjacent g charts. The strongest defensible success would be a reproducible, noise-normalized coupled sector trajectory under intervention. Causal interpretation would still require separating physical coupling from observer knowledge.

10 Conclusion

The numerical and mathematical attacks support Resolution Geometry as a framework for experimental accessibility. In the analyzed optical systems, apparatus changes move stable resolution boundaries; response sectors admit a low-dimensional transport atlas; weak spectral directions remain stable under perturbation; and external controls close exact identifiability holes. These findings justify the map

$$\text{measurement configuration} \longrightarrow \text{admissible observable sector} \longrightarrow \text{transportable recoverable structure.} \quad (27)$$

They do not justify the stronger statement that observation causes a double-slit transition. The double-slit problem enters only as a clearly specified future test: whether path distinguishability and mutual coherence undergo a coupled admissible-sector exchange under one controlled interaction. This separation between demonstrated structure, conjecture, and non-claim is essential to the present result.

Reproducibility

The numerical outputs used in this paper are saved as `rg_wobs_adm_optics_attack.json` and `rg_optics_battery_results.json`. The deterministic random seed is 20260802. Bootstrap counts are 500 per AC load, 1,000 for the spectral perturbation attack, and 2,000 for each optical-activity regression. Input files are the unmodified CSV/XLSX files from the public UniPV Physics Laboratory 3 repository. Any future release should include the analysis script, dependency versions, exact preprocessing rules, and cryptographic hashes of all input files.

References

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